No Uniform Interpolation in Linear Logic

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Abstract

The literature on uniform interpolation in linear logic is very scarce: the sole known result is that the multiplicative-additive fragment of linear logic has the uniform interpolation property, and so does its intuitionistic and affine variants [1]. We exhibit a simple formula of the unit-free multiplicative-exponential fragment of linear logic that cannot have any uniform interpolant in propositional linear logic. This example shows not only that propositional linear logic does not have the uniform interpolation property, but also that its intuitionistic and affine variants neither do. Furthermore, building on this example, we show the undecidability of the following problem in both classical and intuitionistic linear logic: given a formula, does it have a uniform interpolant?

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1 Introduction

Looking at known results for interpolation in linear logic and its fragments, a reader can find a modicum of work, chiefly on Craig's and Lyndon's interpolation and with very little on uniform interpolation. *Craig's interpolation* states that if $A \vdash B$ is provable, then there exists a formula C built only from variables occurring in both A and B and such that both $A \vdash C$ and $C \vdash B$ are provable. *Lyndon's interpolation* is a strengthening of this property taking account the polarity of variables. These two properties have been proved in first-order linear logic and studied in several fragments of this logic. Notably, Roorda [22] proved and disproved Lyndon's interpolation in various fragments of propositional linear logic; Brotherston and Rajeev [2] used display logic to obtain Craig's interpolation in the multiplicative and multiplicative-additive fragments of linear logic, as well as in classical logic (they then formalised these results working with Dawson [5]); Strassburger [25] provides in the calculus of structures for multiplicative-exponential linear logic a result corresponding to Craig's interpolation; a proof of Craig's and Lyndon's interpolation in first-order linear logic, along with links to cut-elimination, has been developed by Saurin [23], who then considered with Fiorillo and Osorio-Valencia [7] this question in the setting of the proof-net syntax of multiplicative linear logic. Let us also mention that Rasga, Carnielli and Sernadas have results for variations of exponential-free linear logic using translations between different logics [21], and that Fussner and Santschi [8] researched Craig's interpolation in extensions of classical and intuitionistic propositional linear logic using algebraic techniques.

The literature is even more scarce on uniform interpolation in linear logic. *Uniform interpolation*, introduced by Pitts [20] in the setting of intuitionistic logic, is a stronger version

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of Craig's interpolation where the interpolant depends only on one of the two formulas under consideration. More precisely, a uniform interpolant for a formula A, with respect to a variable X, is a formula C such that:

- the variables of C are included in those of A, except X must not appear in C;
- $A \vdash C$ is provable;
- for all formulas B such that X does not appear in B and $A \vdash B$ is provable, $C \vdash B$ is provable too.

In particular, uniform interpolants are an interpretation of first-order quantifiers in the logic under consideration, since the formula C above behaves like $\exists X A$ (at least for provability purposes). Not every logic has uniform interpolation, for instance the modal logic S4 does not [10] even thought it has Craig's and Lyndon's interpolations [18, 9]. It is easy to check that classical logic has this property: a uniform interpolant for a formula A with respect to X is simply $A[\top/X] \vee A[\bot/X]$, that is the disjunction of substituting X by true or false. The foundational work of Pitts [20] proves that intuitionistic logic has uniform interpolation. The algorithm to compute an interpolant is more complex than in the classical case, and the proof technique is based on a syntactic approach. This method by Pitts was then adapted to many other logics. In particular, it was used by Alizadeh, Derakhshan and Ono [1] to prove uniform interpolation in the Lambek calculus, as well as in classical and intuitionistic multiplicative-additive linear logic – and also in the affine variants of these two. This is, at the best of the author's knowledge, the *sole* result on uniform interpolation in linear logic to this day. Pitts' method does not extend easily to full propositional linear logic because of the contraction rule, that has a premise sequent "bigger" than its conclusion sequent, and that necessitated a particular treatment in Pitts' paper [20] so as to remove it from the calculus.

Nonetheless, since classical logic and intuitionistic logic both have uniform interpolation, one could expect linear logic to have it too. We show it is not the case by exhibiting a counter-example. Actually, we present formulas A and $(B_n)_{n \in \mathbb{N}}$ such that:

1. an atom X appears in A but not in any of the B_n;
2. $A \vdash B_n$ is provable for all number n;
3. for all formulas C such that X does not appear in C, for any number $n \in \mathbb{N}$, if $A \vdash C$ and $C \vdash B_n$ are both provable, then C is of size at least n.

The result immediately follows: A has no uniform interpolant with respect to X, because such an interpolant would have an unbounded size. Furthermore, the counter-example we exhibit is not only one for uniform interpolation in linear logic, but also in its intuitionistic and affine variants. A subtle use of the exponential modality is mandatory inside the formula A, since uniform interpolation holds with no exponential (case of multiplicative-additive linear logic) and with exponentials everywhere (case of classical logic). This complexity of A entails difficulties to prove some results for sequents of the shape $A \vdash C$. In particular, the most challenging part in this paper is the proof of property 3 above, that involves some technical tools among which geometry of interaction.

Once proved that uniform interpolation does not hold in a given logic, a natural follow-up question is: given a formula, does it has a uniform interpolant? We show that this decision problem is undecidable in both classical and intuitionistic linear logic. This is done by reducing provability in these logics to the existence, or not, of a uniform interpolant. Strikingly, this reduction follows by a modification of our counter-example for a formula A with no uniform interpolant, keeping in particular the "shape" of A.

Outline. We first recall the syntax of linear logic in Section 2, along with a definition of uniform interpolation in this setting. Then, we introduce some useful but technical tools in

$$
\frac{}{\vdash X^+, X^-}\ (ax) \qquad \frac{\vdash A, \Gamma \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta}\ (cut)
$$

$$
\frac{\vdash A, B, \Gamma}{\vdash A \parr B, \Gamma}\ (\parr) \qquad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta}\ (\otimes) \qquad \frac{\vdash \Gamma}{\vdash \bot, \Gamma}\ (\bot) \qquad \frac{}{\vdash 1}\ (1)
$$

$$
\frac{\vdash A, \Gamma \quad \vdash B, \Gamma}{\vdash A \,\&\, B, \Gamma}\ (\&) \qquad \frac{\vdash A, \Gamma}{\vdash A \oplus B, \Gamma}\ (\oplus_1) \qquad \frac{\vdash B, \Gamma}{\vdash A \oplus B, \Gamma}\ (\oplus_2) \qquad \frac{}{\vdash \top, \Gamma}\ (\top)
$$

$$
\frac{\vdash A, \Gamma}{\vdash ?A, \Gamma}\ (?d) \qquad \frac{\vdash ?A, ?A, \Gamma}{\vdash ?A, \Gamma}\ (?c) \qquad \frac{\vdash \Gamma}{\vdash ?A, \Gamma}\ (?w) \qquad \frac{\vdash A, ?\Gamma}{\vdash !A, ?\Gamma}\ (!)
$$

Figure 1 Inference Rules of propositional Linear Logic

Section 3. The main part, Section 4, consists of a presentation of our counter-example and a proof it cannot have a uniform interpolant using the tools from the preceding section. It is followed in Section 5 by a proof that our counter-example is also one for uniform interpolation in *intuitionistic* linear logic and in *affine* logic (both classical and intuitionistic). Finally, our counter-example is adapted in Section 6 so as to obtain undecidability of whether a uniform interpolant exists, in linear logic as well as in its intuitionistic variant.

2 Linear Logic & Uniform Interpolation

We present here linear logic in a standard way as a unilateral sequent calculus [11]. In particular, a proof is a *derivation* (tree) whose conclusion sequent is of the shape $\vdash \Gamma$.

Formulas are defined by the following grammar, with X an atom in some countable set:

$$
A, B ::= \underbrace{X^- \mid X^+}_{\text{atom}} \mid \underbrace{A \parr B \mid A \otimes B \mid \bot \mid 1}_{\text{multiplicative}} \mid \underbrace{A \,\&\, B \mid A \oplus B \mid \top \mid 0}_{\text{additive}} \mid \underbrace{?A \mid !A}_{\text{exponential}}
$$

Orthogonality $(\cdot)^\perp$ (*a.k.a.* negation, duality) is an involution defined inductively by:

$$
\begin{array}{llll}
(X^-)^\perp \overset{\text{def}}{=} X^+ & (A \parr B)^\perp \overset{\text{def}}{=} A^\perp \otimes B^\perp & (A \,\&\, B)^\perp \overset{\text{def}}{=} A^\perp \oplus B^\perp & (?A)^\perp \overset{\text{def}}{=} !A^\perp \\
(X^+)^\perp \overset{\text{def}}{=} X^- & (A \otimes B)^\perp \overset{\text{def}}{=} A^\perp \parr B^\perp & (A \oplus B)^\perp \overset{\text{def}}{=} A^\perp \,\&\, B^\perp & (!A)^\perp \overset{\text{def}}{=} ?A^\perp \\
 & \bot^\perp \overset{\text{def}}{=} 1 & \top^\perp \overset{\text{def}}{=} 0 & \\
 & 1^\perp \overset{\text{def}}{=} \bot & 0^\perp \overset{\text{def}}{=} \top &
\end{array}
$$

In addition to the connectives in the above grammar, we define another binary connective known as **linear implication** by $A \multimap B \overset{\text{def}}{=} A^\perp \parr B$. **Rules** are given on Figure 1, where A and B stand for arbitrary formulas, Γ and Δ for sets of (occurrences of) formulas, and $?\Gamma$ for a set of (occurrences of) ?-formulas. Remark we restrict the *ax*-rule to *atomic formulas*, a well-known simplification. This sequent calculus is equipped with a **cut-elimination** procedure, denoted by $\longrightarrow$, that is weakly normalising and that we recall in Appendix A. Let us also mention that linear logic has the **sub-formula property** [11]: given a *cut-free* derivation π of conclusion sequent $\vdash \Gamma$, all the formulas appearing in π are sub-formulas of the formulas in Γ.

Uniform interpolation in linear logic can be stated as follows. We denote by $\mathcal{V}(A)$ the set of atoms occurring in the formula A – *e.g.* $\mathcal{V}((X^+ \parr X^-) \,\&\, Y^-) = \{X, Y\}$.

▶ **Definition 1** (Uniform Interpolant). *Given a formula A and an atom X, a* ***uniform interpolant*** *for A with respect to X is a formula C such that:*

- $\mathcal{V}(C) \subseteq \mathcal{V}(A) \setminus \{X\}$;
- $\vdash A^\perp, C$ *is provable;*

- *for all formulas B, if $X \notin \mathcal{V}(B)$ and $\vdash A^\perp, B$ is provable then $\vdash C^\perp, B$ is provable.*

*The **uniform interpolation property** holds when every formula admits a uniform interpolant with respect to every atom.*

3 Toolbox

We present here two tools used to prove that our counter-example has no uniform interpolant. The first is a variant of *slices*, the second of *Geometry Of Interaction*. They are needed solely to prove our most technical statement in Section 4, corresponding to property 3 in the introduction, as well as the look-alike statement in Section 6.

3.1 A variant of Slices

The &-rule stands apart in linear logic since it is the sole rule with some sharing of the context Γ. A useful tool when considering &-rules is the notion of a *slice* [11, 13] which is a derivation missing some additive component: a slice is obtained from a derivation by deleting one sub-tree of each &-rule. In a slice, no rule duplicates a context. As our framework has not only additive but also exponential rules, the well-behaved concept is the one of a 0-*slice*, meaning we only slice the &-rules that are not above a !-rule, *i.e.* those at depth 0. This notion of a slice already appeared in the literature, see *e.g.* [19].

▶ **Definition 2** (0-slice). *For π a derivation, a* 0***-slice*** *of π is a (usually non-derivation) tree π' obtained by deleting one of the two sub-trees of each &-rule of π* that is not above a !-rule. *Thus, in π' some (but maybe not all) &-rules are unary:*

$$\dfrac{\vdash A, \Gamma}{\vdash A \,\&\, B, \Gamma}\,(\&_1) \qquad \dfrac{\vdash B, \Gamma}{\vdash A \,\&\, B, \Gamma}\,(\&_2)$$

▶ **Example 3.** Here is a derivation followed by its two 0-slices (note it has three slices):

$$\dfrac{\dfrac{}{\vdash \top, !(1 \,\&\, 1)}\,(\top) \quad \dfrac{\dfrac{\dfrac{\dfrac{}{\vdash 1}\,(1) \quad \dfrac{}{\vdash 1}\,(1)}{\vdash 1 \,\&\, 1}\,(\&)}{\vdash !(1 \,\&\, 1)}\,(!)}{\vdash \bot, !(1 \,\&\, 1)}\,(\bot)}{\vdash \top \,\&\, \bot, !(1 \,\&\, 1)}\,(\&) \qquad \dfrac{\dfrac{}{\vdash \top, !(1 \,\&\, 1)}\,(\top)}{\vdash \top \,\&\, \bot, !(1 \,\&\, 1)}\,(\&_1) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash 1}\,(1) \quad \dfrac{}{\vdash 1}\,(1)}{\vdash 1 \,\&\, 1}\,(\&)}{\vdash !(1 \,\&\, 1)}\,(!)}{\vdash \bot, !(1 \,\&\, 1)}\,(\bot)}{\vdash \top \,\&\, \bot, !(1 \,\&\, 1)}\,(\&_2)$$

Cut-elimination can be defined in the natural way for 0-slices, taking the definition for derivations and extending the $\& - \oplus$ key case with

$$\dfrac{\dfrac{\vdash A, \Gamma}{\vdash A \,\&\, B, \Gamma}\,(\&_1) \quad \dfrac{\vdash A^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta}\,(\oplus_1)}{\vdash \Gamma, \Delta}\,(cut) \quad \longrightarrow \quad \dfrac{\vdash A, \Gamma \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta}\,(cut)$$

$$\dfrac{\dfrac{\vdash B, \Gamma}{\vdash A \,\&\, B, \Gamma}\,(\&_2) \quad \dfrac{\vdash B^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta}\,(\oplus_2)}{\vdash \Gamma, \Delta}\,(cut) \quad \longrightarrow \quad \dfrac{\vdash B, \Gamma \quad \vdash B^\perp, \Delta}{\vdash \Gamma, \Delta}\,(cut)$$

as well as replacing &-rules with $\&_1$- or $\&_2$-rules during a $?d - !$ key case (where a !-rule is removed). Observe it is not possible to eliminate *cut*-rules of the two following shapes:

$$\dfrac{\dfrac{\vdash A, \Gamma}{\vdash A \,\&\, B, \Gamma}\,(\&_1) \quad \dfrac{\vdash B^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta}\,(\oplus_2)}{\vdash \Gamma, \Delta}\,(cut) \qquad \dfrac{\dfrac{\vdash B, \Gamma}{\vdash A \,\&\, B, \Gamma}\,(\&_2) \quad \dfrac{\vdash A^\perp, \Delta}{\vdash A^\perp \oplus B^\perp, \Delta}\,(\oplus_1)}{\vdash \Gamma, \Delta}\,(cut)$$

The only property of 0-slices we need is their behaviour with respect to cut-elimination.

▶ **Lemma 4.** *Let π and ϕ be derivations such that $\pi \longrightarrow \phi$. For each 0-slice ϕ' of ϕ, there exists a 0-slice π' of π such that $\pi' \longrightarrow \phi'$ or $\pi' = \phi'$.*

Proof. We can check that each cut-elimination step respects this property, with the equality case coming from a reduction on rules not in the considered 0-slice. Some cases are detailed in Appendix C. ◀

▶ Remark 5. Observe a corresponding result for slices does not hold, because a formula of the shape $?(A \oplus B)$ can be associated to both a $\oplus_1$- and a $\oplus_2$-rule. This is a phenomenon at play, *e.g.*, in the Seely isomorphism $?(A \oplus B) \simeq ?A \parr ?B$. The cut-elimination case responsible for this failure is the $?c - !$ key case, because the two copies of the duplicated derivation (the "! box") may not delete the same sub-trees for all of their corresponding &-rules.

3.2 A variant of Geometry Of Interaction

We define here the key tool for proving our counter-example has no uniform interpolant, previously used in the restricted setting of unit-free multiplicative-additive linear logic [6]. It falls within the spirit of *Geometry Of Interaction* (shortened as *GOI*) [12]. The quite unusual part is that in the next section we will look at *connectives* of the *cut*-formulas along a path from GOI.

▶ **Definition 6** (GOI projection graph). *The **GOI projection graph** $\mathcal{G}_\pi$ of a derivation, or of a 0-slice, π of conclusion $\vdash \Gamma$ is the undirected edge-bicoloured graph defined as follows, colors being ax and cut.*

Vertices: *All occurrences of atomic formulas X^+ and X^- (for all atoms) from the formulas of Γ and from the cut-formulas A and $A^\perp$ in π, with each cut-rule yielding a pair of cut-formulas giving separate and dual occurrences.*

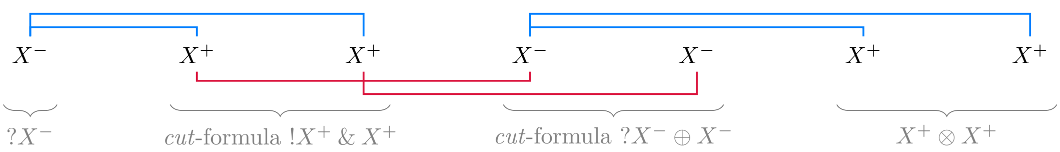
ax-edges: *For each pair of occurrences X^+ and X^- such that π contains an ax-rule $\dfrac{}{\vdash X^+, X^-}\,(ax)$, there is an ax-edge between X^+ and X^-.*

cut-edges: *For each cut-rule $\dfrac{\vdash A, \Delta \qquad \vdash A^\perp, \Sigma}{\vdash \Delta, \Sigma}\,(cut)$ in π, there is a cut-edge between dual occurrences from the two formulas A and $A^\perp$.*

In our figures, ax-edges are coloured in **a**zure and depicted **a**bove the vertices, whereas cut-edges are coloured in **c**rimson and depicted below the vertices.

▶ **Example 7.** Here is a derivation π followed by its GOI projection graph $\mathcal{G}_\pi$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax)}{\vdash ?X^-, X^+}\,(?d)}{\vdash ?X^-, !X^+}\,(!) \qquad \dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax)}{\vdash ?X^-, X^+}\,(?d)}{\vdash ?X^-, !X^+ \,\&\, X^+}\,(\&) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax)}{\vdash ?X^-, X^+}\,(?d) \qquad \dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax)}{\vdash ?X^-, X^+}\,(?d)}{\vdash ?X^-, ?X^-, X^+ \otimes X^+}\,(\otimes)}{\vdash ?X^-, X^+ \otimes X^+}\,(?c)}{\vdash ?X^- \oplus X^-, X^+ \otimes X^+}\,(\oplus_1)}{\vdash ?X^-, X^+ \otimes X^+}\,(cut)$$



▶ Remark 8. Observe they are two ways to obtain several ax-edges on a same occurrence: this occurrence can be duplicated using a $?c$-rule on one of its ancestor (in the formula tree), or by being in the context of a &-rule. Both cases can be observed in Example 7.

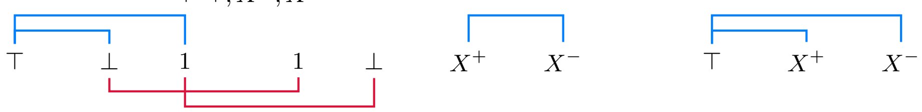
We will consider **alternating** paths in a GOI projection graph, which are paths[1] where consecutive edges are coloured differently – *e.g.* we start with an ax-edge, then take a cut-edge, then an ax-edge, and so on. We have a typical result of GOI about *persistent* paths (see *e.g.* [3]): some paths are preserved through cut-elimination. What is relevant for our purposes is that this holds not only for cut-elimination between derivations but also between 0-slices.

▶ **Lemma 9.** *Consider a cut-elimination step $\pi \longrightarrow \phi$ between two derivations, or between two* 0*-slices, of conclusion $\vdash \Gamma$. Let u and v be occurrences of atoms from Γ. If there is an alternating path between u and v in $\mathcal{G}_\phi$, then there is an alternating path between them in $\mathcal{G}_\pi$.*

Proof. By inspection on the kind of the cut-elimination step. Since we only care about ax-rules and atoms, the non-immediate cases are either trivial with $\mathcal{G}_\phi \subseteq \mathcal{G}_\pi$ ($\& - \oplus$ key cases, $?d - !$ key case, $?w - !$ key case, $\top - cut$ commutative case) or easy to check (ax key case, $?c - !$ key case, $\& - cut$ commutative case). A few of these cases are given in Appendix C. ◀

▶ Remark 10. It is easy to find a counter-example to the reciprocal of Lemma 9. One could be tempted to consider not only ax-rules but also $\top$-rules in a GOI projection graph, also adding vertices for all occurrences of units. Such an altered setting does not suit our purposes for Lemma 9 would not hold – consider the following, with the occurrences of $\top$ and X^+:

$$\dfrac{\dfrac{}{\vdash \top, \bot \parr 1}\,(\top) \qquad \dfrac{\dfrac{}{\vdash 1}\,(1) \quad \dfrac{\dfrac{}{\vdash X^+, X^-}\,(ax)}{\vdash \bot, X^+, X^-}\,(\bot)}{\vdash 1 \otimes \bot, X^+, X^-}\,(\otimes)}{\vdash \top, X^+, X^-}\,(cut) \qquad \longrightarrow \qquad \dfrac{}{\vdash \top, X^+, X^-}\,(\top)$$



4 A Counter-Example to Uniform Interpolation

Let us fix two atoms X and Y. Our counter-example is:

$$A \overset{\text{def}}{=} X^+ \otimes !\big(X^+ \multimap (X^+ \otimes Y^+)\big) \otimes (X^+ \multimap Y^+) \qquad \begin{cases} B_0 & \overset{\text{def}}{=} Y^+ \otimes Y^+ \\ B_{n+1} & \overset{\text{def}}{=} B_n \otimes Y^+ \end{cases} \tag{1}$$

The main intuition is the following. The formula A is made of three parts: X^+, a "reusable machine" taking as input a X^+ and returning a X^+ and a Y^+, and a "one-use machine" taking as input a X^+ and returning a Y^+. Thence, A can "produce" any non-null number of Y^+, so that $A \vdash B_n$ is provable for all $n \in \mathbb{N}$. Therefore, one may think that $!Y^+ \otimes Y^+$ is a uniform interpolant for A, and indeed we do have $!Y^+ \otimes Y^+ \vdash B_n$ provable; however, $A \vdash !Y^+ \otimes Y^+$ cannot be proved. This generalizes to all "uniform interpolant candidates".

[1] A path is a finite alternating sequence of vertices and edges $(v_0, e_1, v_1, e_2, v_2, \ldots, e_n, v_n)$ such that for all $i \in \{1, \ldots, n\}$, the two endpoints of e_i are v_{i-1} and v_i.

Let us point out that A can be viewed as some "pseudo ! exponential" of Y^+, since the following dereliction and selection rules for Y^- are admissible:

$$\dfrac{\vdash Y^-, \Gamma}{\vdash A^\perp, \Gamma} \qquad\qquad \dfrac{\vdash Y^-, A^\perp, \Gamma}{\vdash A^\perp, \Gamma}$$

The formula A is reminiscent of two other "pseudo ! exponentials" of Y^+ from the literature:

- $Y^+ \otimes !(Y^+ \multimap (Y^+ \otimes Y^+)) \otimes !(Y^+ \multimap 1)$ used by Yves Lafont to discuss some non-admissible !-introduction rule [15, page 14];
- $X^+ \otimes !(X^+ \multimap (X^+ \otimes X^+)) \otimes !(X^+ \multimap 1) \otimes !(X^+ \multimap Y^+)$ used by Anupam Das to prove different the exponentials of linear logic and their encoding in multiplicative-additive linear logic with least and greatest fixed points [4] – actually the formula in the reference is not exactly the previous one but an isomorphic formula.

The later formula by Anupam Das could also be used here in place of our A. Nonetheless, we preferred the definition in Equation (1) because it is smaller and we do not need the full power of an exponential – in particular not the $?w$-rule.

To obtain that there is no uniform interpolant for A, it stays to check the three properties claimed in the introduction: $X \notin \mathcal{V}(B_n)$; and $A \vdash B_n$ is provable; and for all formulas C such that $X \notin \mathcal{V}(C)$, if $A \vdash C$ and $C \vdash B_n$ are both provable, then C is of size at least n. Note that the first property holds by definition. The second is easy to prove.

▶ **Lemma 11.** *For all $n \in \mathbb{N}$, there is a derivation of $\vdash A^\perp, B_n$.*

Proof. We build a derivation π_n of $\vdash X^-, ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n$ by induction on $n \in \mathbb{N}$. The result then follows by exhibiting:

$$\dfrac{\dfrac{\overset{\pi_n}{\vdash X^-, ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n}}{\vdash X^-, ?(X^+ \otimes (X^- \parr Y^-)) \parr (X^+ \otimes Y^-), B_n}\,(\parr)}{\vdash X^- \parr ?(X^+ \otimes (X^- \parr Y^-)) \parr (X^+ \otimes Y^-), B_n}\,(\parr)$$

We define the wanted derivations as:

$$\pi_0 \overset{\text{def}}{=} \dfrac{\dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax) \qquad \dfrac{\dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax) \qquad \dfrac{\dfrac{}{\vdash Y^-, Y^+}\,(ax) \quad \dfrac{}{\vdash Y^-, Y^+}\,(ax)}{\vdash Y^-, Y^-, Y^+ \otimes Y^+}\,(\otimes)}{\vdash X^-, Y^-, X^+ \otimes Y^-, Y^+ \otimes Y^+}\,(\otimes)}{\vdash X^- \parr Y^-, X^+ \otimes Y^-, Y^+ \otimes Y^+}\,(\parr)}{\vdash X^-, X^+ \otimes (X^- \parr Y^-), X^+ \otimes Y^-, Y^+ \otimes Y^+}\,(\otimes)}{\vdash X^-, ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, Y^+ \otimes Y^+}\,(?d)$$

$$\pi_{n+1} \overset{\text{def}}{=} \dfrac{\dfrac{\dfrac{\dfrac{}{\vdash X^-, X^+}\,(ax) \qquad \dfrac{\dfrac{\overset{\pi_n}{\vdash ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n} \qquad \dfrac{}{\vdash Y^-, Y^+}\,(ax)}{\vdash X^-, Y^-, ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+}\,(\otimes)}{\vdash X^- \parr Y^-, ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+}\,(\parr)}{\vdash X^-, X^+ \otimes (X^- \parr Y^-), ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+}\,(\otimes)}{\vdash X^-, ?(X^+ \otimes (X^- \parr Y^-)), ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+}\,(?d)}{\vdash X^-, ?(X^+ \otimes (X^- \parr Y^-)), X^+ \otimes Y^-, B_n \otimes Y^+}\,(?c)$$

◀

The third wished property is more complex to prove: we use the tools from Section 3. It also needs an intermediate result, that is not surprising, albeit a little knotty to prove.

▶ **Lemma 12.** *Let C be a formula such that $X \notin \mathcal{V}(C)$ and $\vdash A^\perp, C$ is provable. There exists a cut-free derivation of $\vdash A^\perp, C$ in which no $!$-rule has $?\,(X^+ \otimes (X^- \parr Y^-))$ in its conclusion sequent.*

Proof. Consider a *cut*-free derivation π of $\vdash A^\perp, C$ *with a minimal number of* $!$*-rules having* $?\,(X^+ \otimes (X^- \parr Y^-))$ *in their conclusion sequents.* We prove this number is null.

By the sub-formula property and the constraint on the context in a !-rule, any given !-rule in π is of the shape

$$\dfrac{\begin{array}{c}\phi\\ \vdash \left(?\,(X^+ \otimes (X^- \parr Y^-))\right)^i, D, ?\Delta\end{array}}{\vdash \left(?\,(X^+ \otimes (X^- \parr Y^-))\right)^i, !D, ?\Delta}\;(!)$$

with a conclusion sequent containing $i \geq 0$ copies of $?\,(X^+ \otimes (X^- \parr Y^-))$ for some $i \in \mathbb{N}$, and where $!D, ?\Delta$ are sub-formulas of C. Towards a contradiction, assume $i \geq 1$. Substituting X by 0 in ϕ yields a derivation $\phi[0/X]$ of $\vdash \left(?\,(0 \otimes (\top \parr Y^-))\right)^i, D, ?\Delta$ since, by hypothesis, there is no X in $D, ?\Delta$. Consider any result τ of fully eliminating *cut*-rules in

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash \top, 0 \otimes Y^+}\;(\top)}{\vdash \top \parr (0 \otimes Y^+)}\;(\parr)}{\vdash !\,(\top \parr (0 \otimes Y^+))}\;(!) \qquad \dfrac{\begin{array}{c}\phi[0/X]\\ \vdash \left(?\,(0 \otimes (\top \parr Y^-))\right)^i, D, ?\Delta\end{array}}{\vdash ?\,(0 \otimes (\top \parr Y^-)), D, ?\Delta}\;(?c)}{\vdash D, ?\Delta}\;(cut)}{\vdash !D, ?\Delta}\;(!)}{\vdash \left(?\,(X^+ \otimes (X^- \parr Y^-))\right)^i, !D, ?\Delta}\;(?w)$$

where doubled inference rules mean we apply the rule several times depending on i. Then, τ has no !-rule having $?\,(X^+ \otimes (X^- \parr Y^-))$ in its conclusion sequent by the sub-formula property. Therefore, the derivation obtained from π by substituting with τ the sub-tree

$$\dfrac{\begin{array}{c}\phi\\ \vdash \left(?\,(X^+ \otimes (X^- \parr Y^-))\right)^i, D, ?\Delta\end{array}}{\vdash \left(?\,(X^+ \otimes (X^- \parr Y^-))\right)^i, !D, ?\Delta}\;(!)$$ contradicts the minimality assumption on π. ◀

▶ Remark 13. The above lemma is the part of the demonstration that fails in presence of quantifiers – remember that uniform interpolation is trivial with first-order quantifiers. Indeed, setting $C \stackrel{\text{def}}{=} \exists X A$, the sequent $\vdash A^\perp, C$ is provable but all its *cut*-free derivations contain some !-rule having $?\,(X^+ \otimes (X^- \parr Y^-))$ in its conclusion sequent – simply because this is the only ?-formula and $\vdash !\,(Z^+ \multimap (Z^+ \otimes Y^+))$ is not provable for any atom Z. The reason our proof of Lemma 12 fails with quantifiers is that the sub-formula property no longer holds as stated in this setting: even if $X \notin \mathcal{V}(C)$, an existential quantifier in C may be instantiated with X.

▶ **Proposition 14.** *Let C be a formula such that $X \notin \mathcal{V}(C)$. If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are both provable for some number $n \in \mathbb{N}$, then C contains at least $n+2$ occurrences of Y^+.*

Proof. Let us call π and ϕ derivations of respectively $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$. We assume them *cut*-free without any loss of generality. We will use the GOI projection graph of (some

0-slice of) their composition by *cut* to identify Y^-'s in $C^\perp$ linked to the Y^+'s of B_n. Thanks to alternating paths, we will prove that these occurrences of Y^- are pairwise distinct. This yields one Y^- in $C^\perp$ for each Y^+ in B_n, concluding the proof.

Finding an alternating path in some 0**-slice.** Observe that $\dfrac{\overset{\pi}{\vdash A^\perp, C} \qquad \overset{\phi}{\vdash C^\perp, B_n}}{\vdash A^\perp, B_n}\,(cut)$ reduces to a *cut*-free derivation of $\vdash A^\perp, B_n$, that we call τ. This τ is its own unique 0-slice, since there is no &-connective in $A^\perp$ nor in B_n. By repeated applications of Lemma 4, there is a 0-slice μ of $\dfrac{\overset{\pi}{\vdash A^\perp, C} \qquad \overset{\phi}{\vdash C^\perp, B_n}}{\vdash A^\perp, B_n}\,(cut)$ such that $\mu \longrightarrow^* \tau$.

Consider some occurrence Y^+_α in B_n, with the index α used to distinguish it from other occurrences. Since τ is a *cut*-free derivation of $\vdash A^\perp, B_n$, there must be an *ax*-rule between this occurrence Y^+_α and one of the two occurrences of Y^- in $A^\perp$: call it Y^-_1. Hence, there is an alternating path (made of one *ax*-edge) between these two occurrences in the GOI projection graph $\mathcal{G}_\tau$ of τ. By repeated applications of Lemma 9, there also is an alternating path p between Y^-_1 and Y^+_α in $\mathcal{G}_\mu$. Thus, there is an occurrence Y^-_β and an *ax*-rule $\dfrac{}{\vdash Y^-_\beta, Y^+_\alpha}\,(ax)$ in μ and Y^-_β–Y^+_α is the last edge of p. This occurrence Y^-_β can only be in $C^\perp$.

We use the alternating path p to show there is no sub-formula $?D$ of $C^\perp$ containing Y^-_β – say differently, there is no $?$ between the root of the formula $C^\perp$ and Y^-_β. Once this claim is proved, we are done: each occurrence of Y^+ in B_n is linked by an *ax*-edge to an occurrence of Y^- in $C^\perp$. All these occurrences of Y^- are pairwise distinct (recall Remark 8): there is no sub-formula $?D$ of $C^\perp$ containing such an occurrence, so they cannot be contracted; and there is no &-rule that can share the identified *ax*-rules since we are in a 0-slice, and these *ax*-rules cannot be above a !-rule for there is no ?-connective nor !-connective below the Y^+ of B_n. Therefore, we have found $n+2$ occurrences of Y^- in $C^\perp$, as wished.

Exploiting the alternating path. We prove our claim by showing a stronger result:

- for all occurrences of Y^- in C (respectively in $C^\perp$) belonging to p, there is no $?$ between the root of C (respectively of $C^\perp$) and these occurrences;
- for all occurrences of Y^+ in C (respectively in $C^\perp$) belonging to p, there is no $!$ between the root of C (respectively of $C^\perp$) and these occurrences.

We proceed by following the edges of p starting from its endpoint Y^-_1 in $A^\perp$.

Exploiting the alternating path: base case. Consider the first edge Y^-_1–Y^+_2 of p, with Y^+_2 necessarily in C. Using Lemma 12, without any loss of generality, there is no !-connective between the root of C and Y^+_2. Otherwise, since there is an *ax*-rule on $\vdash Y^-_1, Y^+_2$, there would be a !-rule (associated to this !-connective) with in its conclusion sequent a ?-sub-formula of $A^\perp$ containing Y^-_1, meaning with $?(X^+ \otimes (X^- \parr Y^-))$: contradiction.

Exploiting the alternating path: inductive case. See Figure 2 for an illustration of our present reasoning. Assume the results holds for a prefix of p, and let us consider its last edge.

1. Suppose this prefix ends with an *ax*-edge Y^-_k–Y^+_{k+1} with Y^+_{k+1} in C. We know there is no $!$ between the root of C and Y^+_{k+1}. As p is alternating, the subsequent edge is a *cut*-edge Y^+_{k+1}–Y^-_{k+2} with Y^-_{k+2} in $C^\perp$. Since Y^+_{k+1} and Y^-_{k+2} are dual occurrences by definition of a *cut*-edge, there is no $?$ between the root of $C^\perp$ and Y^-_{k+2}.

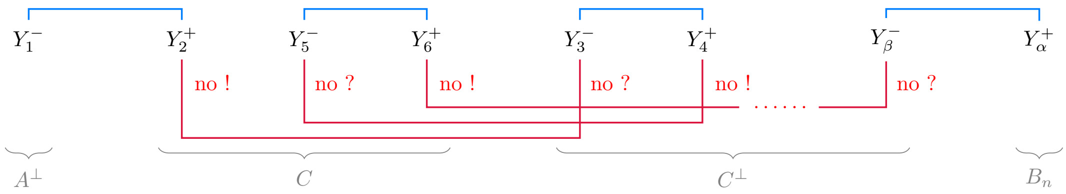


Figure 2 Alternating path p in the GOI projection graph $\mathcal{G}_\mu$ in the proof of Proposition 14

2. Suppose this prefix ends with a *cut*-edge Y_k^+–Y_{k+1}^- with Y_{k+1}^- in $C^\perp$. We know there is no ? between the root of $C^\perp$ and Y_{k+1}^-. As p is alternating, the subsequent edge is an *ax*-edge Y_{k+1}^-–Y_{k+2}^+ with Y_{k+2}^+ either in B_n or in $C^\perp$. In the first case, we are done: we have $Y_{k+2}^+ = Y_\alpha^+$ since there is no *cut*-edge with endpoint Y_{k+2}^+ to continue our path, and p is supposed to end in Y_α^+. In the second case, since there is an *ax*-rule $\dfrac{}{\vdash Y_{k+1}^-, Y_{k+2}^+}\,(ax)$ by definition of an *ax*-edge, it follows there is no ! between the root of $C^\perp$ and Y_{k+2}^+ – because the context condition of a corresponding !-rule would prevent the presence of Y_{k+1}^- in all sequents above it.
3. Suppose this prefix ends with an *ax*-edge Y_k^-–Y_{k+1}^+ with Y_{k+1}^+ in $C^\perp$. This case is identical to case 1 up to switching C and $C^\perp$.
4. Suppose this prefix ends with a *cut*-edge Y_k^+–Y_{k+1}^- with Y_{k+1}^- in C. This case is identical to case 2, with no need to consider Y_{k+2}^+ in $A^\perp$ since $A^\perp$ has no Y^+.

This proves our claim, and concludes the proof. ◀

▶ Remark 15. By going in the other direction of the path p, we can similarly show that:
- for all occurrences of Y^- in C (respectively in $C^\perp$) belonging to p, there is no ! between the root of C (respectively of $C^\perp$) and these occurrences;
- for all occurrences of Y^+ in C (respectively in $C^\perp$) belonging to p, there is no ? between the root of C (respectively of $C^\perp$) and these occurrences.

This means the "useful" part of C as identified by p is exponential-free.

Now that we have all needed properties, we easily obtain our main result.

▶ **Theorem 16.** *Linear logic does not have the uniform interpolation property.*

Proof. The formula A cannot have a uniform interpolant C with respect to X, as such a C would have an unbounded number of occurrences of Y^+ by Lemma 11 and Proposition 14. ◀

5 Adaptation to the Intuitionistic and Affine variants

Our counter-example to the uniform interpolation property in linear logic is also a counter-example to it in *intuitionistic* linear logic [14]. This can be directly obtained through a result of Harold Schellinx [24]. Formulas of propositional intuitionistic linear logic are given by:

$$A, B ::= X^+ \mid A \multimap B \mid A \otimes B \mid 1 \mid A \mathbin{\&} B \mid A \oplus B \mid \top \mid 0 \mid !A$$

▶ **Proposition 17** ([24, Proposition 3.8])**.** *Consider a formula D of intuitionist linear logic which does not contain 0 or does not contain $\multimap$. Then $\vdash D$ is provable in linear logic if and only if it is provable in intuitionistic linear logic.*

▶ **Theorem 18.** *Intuitionistic linear logic does not have the uniform interpolation property.*

Proof. The formula A from Equation (1) has no uniform interpolant with respect to X. Indeed, $\vdash A \multimap B_n$ is provable intuitionistically for all $n \in \mathbb{N}$ using Lemma 11 and Proposition 17 – for 0 is not in $A \multimap B_n$. This yields an intuitionistic derivation of $A \vdash B_n$:

$$\dfrac{\dfrac{\dfrac{}{A \vdash A}\,(ax) \qquad \dfrac{}{B_n \vdash B_n}\,(ax)}{A, A \multimap B_n \vdash B_n}\,(\multimap) \qquad \vdash A \multimap B_n}{A \vdash B_n}\,(cut)$$

Hence, we conclude as in the classical case: an interpolant for A would have an unbounded number of occurrences of Y^+ by Proposition 14 and conservativity – *i.e.* a sequent provable in intuitionistic linear logic is *a fortiori* provable in linear logic. ◀

Our counter-example is also one to the uniform interpolation property in *affine* linear logic, both classical and intuitionistic. Affine logic is defined as linear logic where the $?w$-rule is replaced by the more general $\dfrac{\vdash \Gamma}{\vdash A, \Gamma}\,(w)$. It is equipped with a weakly normalising cut-elimination procedure – see Appendix B. We need here one more intermediate result.

▶ **Lemma 19.** *In affine logic, there is no derivation of* $\Gamma \vdash \bot$ *where* Γ *is a sequent whose formulas are included in* $\{X, !(X \multimap (X \otimes Y)), X \multimap (X \otimes Y), X \otimes Y, Y, X \multimap Y\}$*, with a same formula appearing possibly several times in* Γ*.*

Proof. By contradiction, consider a smallest *cut*-free derivation of $\Gamma \vdash \bot$ with Γ a sequent of the aforementioned shape. Its last rule is either a $\otimes$- $\parr$- $?d$- $?c$- or w-rule. In each case, one of the premises of this rule is a sequent of the aforementioned shape, which contradicts the minimality hypothesis. ◀

▶ **Theorem 20.** *Affine logic and intuitionistic affine logic do not have the uniform interpolation property.*

Proof. We proceed almost exactly as for linear logic, the sole difference being in the beginning of the proof of Proposition 14. There, that in the normalised derivation τ of $\vdash A^\perp, B_n$ (or $A \vdash B_n$) there is an ax-rule on each Y^+ of B_n is no longer immediate: we invoke Lemma 19, which implies no sub-formula of B_n can be weakened in τ – this lemma holding in affine logic, so *a fortiori* in intuitionistic affine logic by conservativity. ◀

6 Undecidability of the existence of a uniform interpolant

A natural question following the failure of the uniform interpolation property is to study the complexity of the subsequent decision problem:

Uniform interpolant: Given a formula A and an atom X, does A have a uniform interpolant with respect to X?

In this section, we show this problem is undecidable in both classical and intuitionistic linear logic. This is done thanks to a reduction to the provability decision problem, better-studied in linear logic and known to be undecidable in both cases [16]:

Provability: Given a formula F, is $\vdash F$ provable?

Our demonstration that Provability can be encoded into Uniform interpolant is very similar to our proof in Section 4 that our counter-example has no uniform interpolant, up to additional technicalities. In particular, it is done by adapting our counter-example of

Equation (1) by adding to it the formula under consideration. Our key result in this section is the one below, given an arbitrary formula F:

$\vdash F$ is provable if and only if the following has a uniform interpolant with respect to X:

$$A \overset{\text{def}}{=} X^+ \otimes ! \big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

Notice how this new A is obtained from the one in Equation (1) by the substitution $Y^+ \mapsto (Y^+ \multimap (Y^+ \otimes !F))$. Since a big part of the reasoning underlying the demonstration of the above result is similar to what has been done in Section 4, for the sake of clarity the corresponding proofs are given in Appendix D. Nonetheless, there are two key differences compared to the previous sections:

- in case $\vdash F$ is provable, we exhibit as uniform interpolant $C \overset{\text{def}}{=} Y^+ \multimap (Y^+ \otimes !F)$;
- since our new formula A to interpolate is more complex than the previous one, we need some more technical results.

The next lemma is an intermediate result for the case where F is provable, used to prove that a certain formula C is a uniform interpolant for A.

▶ **Lemma 21.** *Let F and B be formulas, and X and Y be atoms such that $X \notin \mathcal{V}(F)$ and $X \notin \mathcal{V}(B)$. Pose:*

$$A \overset{\text{def}}{=} X^+ \otimes ! \big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

$$C \overset{\text{def}}{=} Y^+ \multimap (Y^+ \otimes !F)$$

If $\vdash F$ and $\vdash A^\perp, B$ are both provable, then $\vdash C^\perp, B$ is also provable.

Proof. By hypothesis, there exists a derivation ϕ_1 of conclusion sequent $\vdash A^\perp, B$. Let us find a derivation of $\vdash C^\perp, B$. We proceed by transforming ϕ_1 several times.

We begin by modifying ϕ_1 as follows. We pose

$$A' \overset{\text{def}}{=} X^+ \otimes ! \big(X^+ \multimap (X^+ \otimes 1)\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

that is A where the first of the two occurrences of $Y^+ \multimap (Y^+ \otimes !F)$ has been replaced by 1. We obtain a derivation ϕ_2 of $\vdash (A')^\perp, B$ by replacing in ϕ_1 each eventual use of the deleted occurrence $(Y^+ \multimap (Y^+ \otimes !F))^\perp = Y^+ \otimes (Y^- \parr ?F^\perp)$ by the following tree, that replaces back $(1)^\perp = \perp$ with the lacking formula:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash Y^-, Y^+}\,(ax) \quad \dfrac{\begin{matrix}\pi\\ \vdash F\end{matrix}}{\vdash !F}\,(!)}{\vdash Y^-, Y^+ \otimes !F}\,(\otimes)}{\vdash Y^- \parr (Y^+ \otimes !F)}\,(\parr) \qquad \begin{matrix}\vdots\\ \vdash \Gamma, Y^+ \otimes (Y^- \parr ?F^\perp)\end{matrix}}{\dfrac{\vdash \Gamma}{\begin{matrix}\vdash \Gamma, \perp\\ \vdots\end{matrix}}\,(\perp)}\,(cut)$$

where π is a derivation of conclusion $\vdash F$, using that F is assumed provable.

Then, we substitute X by 1 in ϕ_2, obtaining a derivation ϕ_3 of conclusion sequent $\vdash \perp \parr ?(1 \otimes (\perp \parr \perp)) \parr (1 \otimes (Y^+ \otimes (Y^- \parr ?F^\perp))), B$ – remember that $X \notin \mathcal{V}(F) \cup \mathcal{V}(B)$. Now, thanks to the isomorphisms $1 \otimes D \dashv\vdash D$ and $\perp \parr D \dashv\vdash D$, this means there exists a derivation ϕ_4 of conclusion $\vdash ?\bot \parr (Y^+ \otimes (Y^- \parr ?F^\perp)), B$. Since we have the equiprovability $?\bot \dashv\vdash \bot$, and reusing the isomorphism $\bot \parr D \dashv\vdash D$, we finally obtain a derivation ϕ_5 of conclusion $\vdash Y^+ \otimes (Y^- \parr ?F^\perp), B$, that is of $\vdash C^\perp, B$. ◀

The two following lemmas are analogous to previous results from Section 4, namely Lemma 11 and Proposition 14.

▶ **Lemma 22.** *Let F be a formula, and X and Y be atoms. Set:*

$$A \overset{def}{=} X^+ \otimes !\big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

$$\begin{cases} B_0 & \overset{def}{=} (Y^+ \multimap (Y^+ \otimes !F)) \otimes (Y^+ \multimap (Y^+ \otimes !F)) \\ B_{n+1} & \overset{def}{=} B_n \otimes (Y^+ \multimap (Y^+ \otimes !F)) \end{cases}$$

For all $n \in \mathbb{N}$, there is a derivation of $\vdash A^\perp, B_n$.

Proof. It suffices to take the derivations given by Lemma 11 and to apply the following substitution: $Y^+ \mapsto (Y^+ \multimap (Y^+ \otimes !F))$. ◀

▶ **Proposition 23.** *Let F and C be formulas, and X and Y be atoms such that $X, Y \notin \mathcal{V}(F)$ and $X \notin \mathcal{V}(C)$. Assume that F is not provable, and set:*

$$A \overset{def}{=} X^+ \otimes !\big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

$$\begin{cases} B_0 & \overset{def}{=} (Y^+ \multimap (Y^+ \otimes !F)) \otimes (Y^+ \multimap (Y^+ \otimes !F)) \\ B_{n+1} & \overset{def}{=} B_n \otimes (Y^+ \multimap (Y^+ \otimes !F)) \end{cases}$$

If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are both provable for some number $n \in \mathbb{N}$, then C contains at least $n+2$ occurrences of Y^+.

Proof. Mostly like the proof of Proposition 14, with additional technicalities. A demonstration is available in Appendix D. ◀

▶ **Proposition 24.** *Let F be a formula of linear logic. Fixing two atoms $X, Y \notin \mathcal{V}(F)$, $\vdash F$ is provable if and only if the following formula has a uniform interpolant with respect to X:*

$$A \overset{def}{=} X^+ \otimes !\big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

Proof. We proceed according to the provability of F.

Assume that F is provable. We claim that $C \overset{\text{def}}{=} Y^+ \multimap (Y^+ \otimes !F)$ is a uniform interpolant for A with respect to X. Observe that $\mathcal{V}(C) \subseteq \mathcal{V}(A) \setminus \{X\}$, as required.

Here is a derivation of conclusion $\vdash A^\perp, C$:

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{(ax)}
\UnaryInfC{$\vdash X^-, X^+$}
\AxiomC{}
\RightLabel{(ax)}
\UnaryInfC{$\vdash Y^-, Y^+$}
\AxiomC{}
\RightLabel{(ax)}
\UnaryInfC{$\vdash Y^-, Y^+$}
\AxiomC{}
\RightLabel{(ax)}
\UnaryInfC{$\vdash F, F^\perp$}
\RightLabel{$(?d)$}
\UnaryInfC{$\vdash F, ?F^\perp$}
\RightLabel{$(!)$}
\UnaryInfC{$\vdash !F, ?F^\perp$}
\RightLabel{$(\otimes)$}
\BinaryInfC{$\vdash Y^-, ?F^\perp, Y^+ \otimes !F$}
\RightLabel{$(\parr)$}
\UnaryInfC{$\vdash Y^- \parr ?F^\perp, Y^+ \otimes !F$}
\RightLabel{$(\otimes)$}
\BinaryInfC{$\vdash Y^+ \otimes (Y^- \parr ?F^\perp), Y^-, Y^+ \otimes !F$}
\RightLabel{$(\otimes)$}
\BinaryInfC{$\vdash X^-, X^+ \otimes (Y^+ \otimes (Y^- \parr ?F^\perp)), Y^-, Y^+ \otimes !F$}
\RightLabel{$(\parr)$}
\UnaryInfC{$\vdash X^-, X^+ \otimes (Y^+ \otimes (Y^- \parr ?F^\perp)), Y^- \parr (Y^+ \otimes !F)$}
\RightLabel{$(?w)$}
\UnaryInfC{$\vdash X^-, ?\left(X^+ \otimes (X^- \parr (Y^+ \otimes (Y^- \parr ?F^\perp)))\right), X^+ \otimes (Y^+ \otimes (Y^- \parr ?F^\perp)), Y^- \parr (Y^+ \otimes !F)$}
\RightLabel{$(\parr)$}
\UnaryInfC{$\vdash X^-, ?\left(X^+ \otimes (X^- \parr (Y^+ \otimes (Y^- \parr ?F^\perp)))\right) \parr \left(X^+ \otimes (Y^+ \otimes (Y^- \parr ?F^\perp))\right), Y^- \parr (Y^+ \otimes !F)$}
\RightLabel{$(\parr)$}
\UnaryInfC{$\vdash X^- \parr ?\left(X^+ \otimes (X^- \parr (Y^+ \otimes (Y^- \parr ?F^\perp)))\right) \parr \left(X^+ \otimes (Y^+ \otimes (Y^- \parr ?F^\perp))\right), Y^- \parr (Y^+ \otimes !F)$}
\end{prooftree}
$$

Let us fix a formula B such that $X \notin \mathcal{V}(B)$ and there exists a derivation ϕ of conclusion sequent $\vdash A^\perp, B$. Thanks to Lemma 21, there exists a derivation of conclusion $\vdash C^\perp, B$.

Therefore, C is a uniform interpolant for A with respect to X, as claimed.

Assume that F is not provable. We claim that A has no uniform interpolant with respect to X. Let us pose:

$$
\begin{cases}
B_0 & \stackrel{\text{def}}{=} (Y^+ \multimap (Y^+ \otimes !F)) \otimes (Y^+ \multimap (Y^+ \otimes !F)) \\
B_{n+1} & \stackrel{\text{def}}{=} B_n \otimes (Y^+ \multimap (Y^+ \otimes !F))
\end{cases}
$$

Observe that $X \notin \mathcal{V}(B_n)$. Using Lemma 22, $\vdash A^\perp, B_n$ is provable for all $n \in \mathbb{N}$. By Proposition 23, for any formula C such that $X \notin \mathcal{V}(C)$, if $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are both provable, then C contains at least $n + 2$ occurrences of Y^+. Hence, A cannot have any uniform interpolant with respect to X. This concludes the proof. ◀

▶ **Theorem 25.** *In linear logic, the Uniform interpolant problem is undecidable.*

Proof. By Proposition 24, decidability of Uniform interpolant entails decidability of Provability. However, Provability is undecidable in propositional linear logic [16]. ◀

As was the case in Section 5 for the failure of the uniform interpolation property, it is possible to adapt our undecidability result to the *intuitionistic* case.

▶ **Proposition 26.** *Consider the framework of* intuitionistic *linear logic. Let F be a formula. Fixing two atoms $X, Y \notin \mathcal{V}(F)$, $\vdash F$ is provable if and only if the following formula has a uniform interpolant with respect to X:*

$$A \stackrel{def}{=} X^+ \otimes !\left(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\right) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

Proof. One can proceed as in the proof of Proposition 24, using conservativity and that the derivations built in the proofs of Proposition 24 and its intermediate lemmas can be framed as derivations in intuitionistic linear logic instead of classical linear logic. ◀

▶ **Theorem 27.** *In intuitionistic linear logic, the Uniform interpolant problem is undecidable.*

Proof. By Proposition 26, decidability of Uniform interpolant entails decidability of Provability. However, Provability is undecidable in intuitionistic propositional linear logic [16]. ◀

7 Conclusion

We presented a formula of unit-free multiplicative-exponential linear logic that has no uniform interpolant in linear logic, both classical and intuitionistic, as well as in classical and intuitionistic affine logic. This contrasts sharply with the situation of classical logic and intuitionistic logic and multiplicative-additive linear logic, that all have the uniform interpolation property. Our counter-example, *c.f.* Equation (1), sheds some light on when the contraction rule prevents uniform interpolation: it may do so when the formula A to interpolate contains both a non-contractible sub-formula (here X^+) and a contractible sub-formula (here $!(X^+ \multimap (X^+ \otimes Y^+))$), which is impossible in the three aforementioned logics but possible in linear logic and affine logic! Furthermore, please observe that our demonstration also holds in a variant of linear logic with no $?w$-rule and/or with the $?c$-rule replaced by the weaker selection rule $\dfrac{\vdash A, ?A, \Gamma}{\vdash ?A, \Gamma}$. This makes quite clear that what prevents uniform interpolation in linear logic is a contraction rule available only on some formulas (along with the constraint on the context in a !-rule).

The formulas A and B_n we exhibited in Equation (1) are not overly complex, but proving no formula C can be a uniform interpolant was quite tedious as it depends heavily on the necessary structure of C. Geometry of interaction and slices were very useful tools to this end – and we do not do how to prove Proposition 14 without GOI projection graphs.

Considering the uniform interpolation property in linear logic and its fragments, since it trivially holds in the presence of quantifiers (setting $C = \exists X A$ as an interpolant for A with respect to X), and it has been proved in multiplicative-additive linear logic [1] and now been disproved for some formula of multiplicative-exponential linear logic (even allowing the interpolant to use connectives from propositional linear logic), the principal fragment where the question stays open is the (unusual) additive-exponential linear logic. Since in this logic the contraction rule is of limited interest due to the lack of multiplicative connectives, we conjecture the method of Pitts [20] could be adapted to prove uniform interpolation in this fragment, as well as in its intuitionistic and affine variants.

A natural question following the failure of the uniform interpolation property is the subsequent decision problem: given a formula A, does it have a uniform interpolant? We showed this problem to be undecidable in both classical and intuitionistic linear logic, through a reduction to the problem of provability and still exploiting our counter-example for uniform interpolation. Our reasoning might perhaps be also adapted to classical and intuitionistic affine logics, but the provability of these logics is decidable, hence we would only get a lower bound for our problem. Furthermore, this decision problem can also be considered for formulas of fragments of (intuitionistic) linear logic, *e.g.* of multiplicative-exponential (intuitionistic) linear logic. It can be easily checked that our demonstration in Section 6 also applies in the restricted setting of multiplicative-exponential linear logic, so that deciding whether a formula of multiplicative-exponential linear logic has a uniform interpolant is at least harder than provability in this fragment – which is an open problem.

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A Cut-Elimination in Linear Logic

Cut-elimination in linear logic is defined on Tables 1 and 2. It is weakly normalising [11].

ax	$\dfrac{\dfrac{}{\vdash A^\perp, A}(ax) \quad \overset{\pi}{\vdash A,\Gamma}}{\vdash A,\Gamma}(\mathit{cut})$	$\longrightarrow$	$\overset{\pi}{\vdash A,\Gamma}$
$\parr-\otimes-1$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B^\perp, \Gamma}}{\vdash A^\perp \parr B^\perp, \Gamma}(\parr) \quad \dfrac{\overset{\phi}{\vdash A,\Delta} \quad \overset{\tau}{\vdash B,\Sigma}}{\vdash A\otimes B, \Delta,\Sigma}(\otimes)}{\vdash \Gamma,\Delta,\Sigma}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp,B^\perp,\Gamma} \quad \overset{\tau}{\vdash B,\Sigma}}{\vdash A^\perp,\Gamma,\Sigma}(\mathit{cut}) \quad \overset{\phi}{\vdash A,\Delta}}{\vdash\Gamma,\Delta,\Sigma}(\mathit{cut})$
$\parr-\otimes-2$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B^\perp, \Gamma}}{\vdash A^\perp \parr B^\perp, \Gamma}(\parr) \quad \dfrac{\overset{\phi}{\vdash A,\Delta} \quad \overset{\tau}{\vdash B,\Sigma}}{\vdash A\otimes B, \Delta,\Sigma}(\otimes)}{\vdash \Gamma,\Delta,\Sigma}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp,B^\perp,\Gamma} \quad \overset{\phi}{\vdash A,\Delta}}{\vdash B^\perp,\Gamma,\Sigma}(\mathit{cut}) \quad \overset{\tau}{\vdash B,\Sigma}}{\vdash\Gamma,\Delta,\Sigma}(\mathit{cut})$
$\bot-1$	$\dfrac{\dfrac{\overset{\pi}{\vdash\Gamma}}{\vdash\bot,\Gamma}(\bot)\quad \dfrac{}{\vdash 1}(1)}{\vdash\Gamma}(\mathit{cut})$	$\longrightarrow$	$\overset{\pi}{\vdash\Gamma}$
$\&-\oplus_1$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp,\Gamma}\quad\overset{\phi}{\vdash B^\perp,\Gamma}}{\vdash A^\perp \& B^\perp,\Gamma}(\&) \quad \dfrac{\overset{\tau}{\vdash A,\Delta}}{\vdash A\oplus B,\Delta}(\oplus_1)}{\vdash\Gamma,\Delta}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\overset{\pi}{\vdash A^\perp,\Gamma}\quad\overset{\tau}{\vdash A,\Delta}}{\vdash\Gamma,\Delta}(\mathit{cut})$
$\&-\oplus_2$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp,\Gamma}\quad\overset{\phi}{\vdash B^\perp,\Gamma}}{\vdash A^\perp \& B^\perp,\Gamma}(\&) \quad \dfrac{\overset{\tau}{\vdash B,\Delta}}{\vdash A\oplus B,\Delta}(\oplus_2)}{\vdash\Gamma,\Delta}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\overset{\phi}{\vdash B^\perp,\Gamma}\quad\overset{\tau}{\vdash B,\Delta}}{\vdash\Gamma,\Delta}(\mathit{cut})$
$?d-!$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp,\Gamma}}{\vdash ?A^\perp,\Gamma}(?d)\quad\dfrac{\overset{\phi}{\vdash A,?\Delta}}{\vdash !A,?\Delta}(!)}{\vdash\Gamma,?\Delta}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\overset{\pi}{\vdash A^\perp,\Gamma}\quad\overset{\phi}{\vdash A,?\Delta}}{\vdash\Gamma,?\Delta}(\mathit{cut})$
$?c-!$	$\dfrac{\dfrac{\overset{\pi}{\vdash ?A^\perp,?A^\perp,\Gamma}}{\vdash ?A^\perp,\Gamma}(?c)\quad \dfrac{\overset{\phi}{\vdash A,?\Delta}}{\vdash !A,?\Delta}(!)}{\vdash\Gamma,?\Delta}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\dfrac{\dfrac{\overset{\pi}{\vdash ?A^\perp,?A^\perp,\Gamma}\quad\dfrac{\overset{\phi}{\vdash A,?\Delta}}{\vdash !A,?\Delta}(!)}{\vdash ?A^\perp,\Gamma,?\Delta}(\mathit{cut})\quad\dfrac{\overset{\phi}{\vdash A,?\Delta}}{\vdash !A,?\Delta}(!)}{\vdash\Gamma,?\Delta,?\Delta}(\mathit{cut})}{\vdash \Gamma,?\Delta}(?c)$
$?w-!$	$\dfrac{\dfrac{\overset{\pi}{\vdash\Gamma}}{\vdash ?A^\perp,\Gamma}(?w)\quad\dfrac{\overset{\phi}{\vdash A,?\Delta}}{\vdash !A,?\Delta}(!)}{\vdash\Gamma,?\Delta}(\mathit{cut})$	$\longrightarrow$	$\dfrac{\overset{\pi}{\vdash\Gamma}}{\vdash\Gamma,?\Delta}(?w)$

Table 1 Cut-elimination in Linear Logic – Key cases

$cut - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B^\perp, \Gamma, \Delta}\,(cut) \quad \overset{\tau}{\vdash B, \Sigma}}{\vdash \Gamma, \Delta, \Sigma}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B^\perp, \Gamma} \quad \overset{\tau}{\vdash B, \Sigma}}{\vdash A^\perp, \Gamma, \Sigma}\,(cut) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash \Gamma, \Delta, \Sigma}\,(cut)$
$\parr - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, C, \Gamma}}{\vdash A^\perp, B \parr C, \Gamma}\,(\parr) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B \parr C, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, C, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B, C, \Gamma, \Delta}\,(cut)}{\vdash B \parr C, \Gamma, \Delta}\,(\parr)$
$\otimes - cut - 1$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma} \quad \overset{\phi}{\vdash C, \Delta}}{\vdash A^\perp, B \otimes C, \Gamma, \Delta}\,(\otimes) \quad \overset{\tau}{\vdash A, \Sigma}}{\vdash B \otimes C, \Gamma, \Delta, \Sigma}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma} \quad \overset{\tau}{\vdash A, \Sigma}}{\vdash B, \Gamma, \Sigma}\,(cut) \quad \overset{\phi}{\vdash C, \Delta}}{\vdash B \otimes C, \Gamma, \Delta, \Sigma}\,(\otimes)$
$\otimes - cut - 2$	$\dfrac{\dfrac{\overset{\pi}{\vdash B, \Gamma} \quad \overset{\phi}{\vdash A^\perp, C, \Delta}}{\vdash A^\perp, B \otimes C, \Gamma, \Delta}\,(\otimes) \quad \overset{\tau}{\vdash A, \Sigma}}{\vdash B \otimes C, \Gamma, \Delta, \Sigma}\,(cut)$	$\longrightarrow$	$\dfrac{\overset{\pi}{\vdash B, \Gamma} \quad \dfrac{\overset{\phi}{\vdash A^\perp, C, \Delta} \quad \overset{\tau}{\vdash A, \Sigma}}{\vdash C, \Delta, \Sigma}\,(cut)}{\vdash B \otimes C, \Gamma, \Delta, \Sigma}\,(\otimes)$
$\perp - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, \Gamma}}{\vdash A^\perp, \perp, \Gamma}\,(\perp) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash \perp, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash \Gamma, \Delta}\,(cut)}{\vdash \perp, \Gamma, \Delta}\,(\perp)$
$\& - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma} \quad \overset{\phi}{\vdash A^\perp, C, \Gamma}}{\vdash A^\perp, B \,\&\, C, \Gamma}\,(\&) \quad \overset{\tau}{\vdash A, \Delta}}{\vdash B \,\&\, C, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma} \quad \overset{\tau}{\vdash A, \Delta}}{\vdash B, \Gamma, \Delta}\,(cut) \quad \dfrac{\overset{\phi}{\vdash A^\perp, C, \Gamma} \quad \overset{\tau}{\vdash A, \Delta}}{\vdash C, \Gamma, \Delta}\,(cut)}{\vdash B \,\&\, C, \Gamma, \Delta}\,(\&)$
$\oplus_1 - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma}}{\vdash A^\perp, B \oplus C, \Gamma}\,(\oplus_1) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B \oplus C, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B, \Gamma, \Delta}\,(cut)}{\vdash B \oplus C, \Gamma, \Delta}\,(\oplus_1)$
$\oplus_2 - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, C, \Gamma}}{\vdash A^\perp, B \oplus C, \Gamma}\,(\oplus_2) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B \oplus C, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, C, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash C, \Gamma, \Delta}\,(cut)}{\vdash B \oplus C, \Gamma, \Delta}\,(\oplus_2)$
$\top - cut$	$\dfrac{\dfrac{}{\vdash A^\perp, \top, \Gamma}\,(\top) \quad \overset{\pi}{\vdash A, \Delta}}{\vdash \top, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{}{\vdash \top, \Gamma, \Delta}\,(\top)$
$?d - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma}}{\vdash A^\perp, ?B, \Gamma}\,(?d) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash ?B, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash B, \Gamma, \Delta}\,(cut)}{\vdash ?B, \Gamma, \Delta}\,(?d)$
$?c - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, ?B, ?B, \Gamma}}{\vdash A^\perp, ?B, \Gamma}\,(?c) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash ?B, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, ?B, ?B, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash ?B, ?B, \Gamma, \Delta}\,(cut)}{\vdash ?B, \Gamma, \Delta}\,(?c)$
$?w - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, \Gamma}}{\vdash A^\perp, ?B, \Gamma}\,(?w) \quad \overset{\phi}{\vdash A, \Delta}}{\vdash ?B, \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, \Gamma} \quad \overset{\phi}{\vdash A, \Delta}}{\vdash \Gamma, \Delta}\,(cut)}{\vdash ?B, \Gamma, \Delta}\,(?w)$
$! - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash ?A^\perp, B, ?\Gamma}}{\vdash ?A^\perp, !B, ?\Gamma}\,(!) \quad \dfrac{\overset{\phi}{\vdash A, ?\Delta}}{\vdash !A, ?\Delta}\,(!)}{\vdash !B, ?\Gamma, ?\Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash ?A^\perp, B, ?\Gamma} \quad \dfrac{\overset{\phi}{\vdash A, ?\Delta}}{\vdash !A, ?\Delta}\,(!)}{\vdash B, ?\Gamma, ?\Delta}\,(cut)}{\vdash !B, ?\Gamma, ?\Delta}\,(!)$

Table 2 Cut-elimination in Linear Logic – Commutative cases

B Cut-Elimination in Affine Logic

Cut-elimination in affine logic is defined as in linear logic, except that the $?w - !$ key case and the $?w - cut$ commutative case are replaced by the two cases on Table 3. That cut-elimination is weakly normalising in affine logic can be proved similarly to weak normalisation of cut-elimination in linear logic – it is *e.g.* a simple adaptation of the proof in [17].

w	$\dfrac{\dfrac{\overset{\pi}{\vdash \Gamma}}{\vdash A, \Gamma}\,(w) \qquad \vdash A^{\perp}, \Delta}{\vdash \Gamma, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\overset{\pi}{\vdash \Gamma}}{\vdash \Gamma, \Delta}\,(w)$
$w - cut$	$\dfrac{\dfrac{\overset{\pi}{\vdash A, \Gamma}}{\vdash A, B, \Gamma}\,(w) \qquad \overset{\phi}{\vdash A^{\perp}, \Delta}}{\vdash \Gamma, B, \Delta}\,(cut)$	$\longrightarrow$	$\dfrac{\dfrac{\overset{\pi}{\vdash A, \Gamma} \qquad \overset{\phi}{\vdash A^{\perp}, \Delta}}{\vdash \Gamma, \Delta}\,(cut)}{\vdash \Gamma, B, \Delta}\,(w)$

Table 3 Cut-elimination in Affine Logic – Different cases with respect to Linear Logic

C Simple Proofs from Section 3

We provide here some cases for the easy (but tedious) proofs by case study of Lemmas 4 and 9 stated in Section 3.

▶ **Lemma 4.** *Let π and ϕ be derivations such that $\pi \longrightarrow \phi$. For each 0-slice ϕ' of ϕ, there exists a 0-slice π' of π such that $\pi' \longrightarrow \phi'$ or $\pi' = \phi'$.*

Proof. We distinguish cases according to the kind of the cut-elimination step. We present here only a few relevant cases, the others being trivial or similar. Precisely, we cover the $\& - \oplus_1$, $?d - !$ and $?c - !$ key cases, and the $\top - cut$ commutative case.

$\& - \oplus_1$ **key case.** We have

$$\begin{array}{c}\dfrac{\dfrac{\overset{\pi}{\vdash A^{\perp}, \Gamma} \qquad \overset{\phi}{\vdash B^{\perp}, \Gamma}}{\vdash A^{\perp} \& B^{\perp}, \Gamma}\,(\&) \qquad \dfrac{\overset{\tau}{\vdash A, \Delta}}{\vdash A \oplus B, \Delta}\,(\oplus_1)}{\vdash \Gamma, \Delta}\,(cut) \\ \vdots \\ \vdash \Sigma\end{array} \quad \longrightarrow \quad \begin{array}{c}\dfrac{\overset{\pi}{\vdash A^{\perp}, \Gamma} \qquad \overset{\tau}{\vdash A, \Delta}}{\vdash \Gamma, \Delta}\,(cut) \\ \vdots \\ \vdash \Sigma\end{array}$$

and are given a 0-slice s of the reduced $\begin{array}{c}\dfrac{\overset{\pi}{\vdash A^{\perp}, \Gamma} \qquad \overset{\tau}{\vdash A, \Delta}}{\vdash \Gamma, \Delta}\,(cut) \\ \vdots \\ \vdash \Sigma\end{array}$. If the *cut*-rule of this cut-elimination step is not in s, then s is also a 0-slice of the redex. Otherwise, we define the 0-slice r of the redex as making the same choices of sub-trees as in s, except for the $\&$-rule erased during the cut-elimination step where we keep the left sub-tree $\dfrac{\overset{\pi}{\vdash A^{\perp}, \Gamma}}{\vdash A^{\perp} \& B^{\perp}, \Gamma}\,(\&_1)$ so that $r \longrightarrow s$.

$?d - !$ **key case.** Given a 0-slice s of the reduced, we define a 0-slice r of the redex as making the same choice of sub-trees as in s, except for &-rules that are above a !-rule in the redex but not in the reduced, for which there is no choice to make.

$?c - !$ **key case.** This case is trivial solely because a 0-slice leaves the &-rules above a !-rule as they are: we make the same choices for the wanted 0-slice of the redex as the choices made in the given 0-slice of the reduced.

$\top - cut$ **commutative case.** We have

$$\dfrac{\dfrac{}{\vdash A^\perp, \top, \Gamma}(\top) \qquad \overset{\pi}{\vdash A, \Delta}}{\dfrac{\vdash \top, \Gamma, \Delta}{\begin{array}{c}\vdots\\ \vdash \Sigma\end{array}}}(cut) \quad \longrightarrow \quad \begin{array}{c}\dfrac{}{\vdash \top, \Gamma, \Delta}(\top)\\ \vdots\\ \vdash \Sigma\end{array}$$

and are given a 0-slice s of the reduced $\begin{array}{c}\dfrac{}{\vdash \top, \Gamma, \Delta}(\top)\\ \vdots\\ \vdash \Sigma\end{array}$. If the $\top$-rule of this cut-elimination step is not in s, then s is also a 0-slice of the redex. Otherwise, we define the 0-slice r of the redex as making the same choices of sub-trees as in s, except for the &-rules of π erased during the cut-elimination step where we keep an arbitrary sub-tree, say always the left one, so that $r \longrightarrow s$. ◀

▶ **Lemma 9.** *Consider a cut-elimination step $\pi \longrightarrow \phi$ between two derivations, or between two 0-slices, of conclusion $\vdash \Gamma$. Let u and v be occurrences of atoms from Γ. If there is an alternating path between u and v in $\mathcal{G}_\phi$, then there is an alternating path between them in $\mathcal{G}_\pi$.*

Proof. We distinguish cases according to the kind of the cut-elimination step. We present here only the ax and $?c - !$ key cases, the others being trivial or similar.

ax **key case.** We have

$$\pi = \dfrac{\dfrac{}{\vdash X^+, X^-}(ax) \qquad \overset{\tau}{\vdash X^+, \Delta}}{\dfrac{\vdash X^+, \Delta}{\begin{array}{c}\vdots\\ \vdash \Gamma\end{array}}}(cut) \quad \longrightarrow \quad \begin{array}{c}\overset{\tau}{\vdash X^+, \Delta}\\ \vdots\\ \vdash \Gamma\end{array} = \phi$$

with an alternating path p between u and v in the GOI projection graph $\mathcal{G}_\phi$. Let us call x the vertex of $\mathcal{G}_\pi$ corresponding to X^+ in the sequent $\vdash X^+, \Delta$ below the cut-rule, and x' and x'' the vertices corresponding respectively to the cut-formulas X^- and X^+.

We observe that the vertices of $\mathcal{G}_\phi$ are those of $\mathcal{G}_\pi$ except for x' and x''. Moreover, the edges of $\mathcal{G}_\phi$ are those of $\mathcal{G}_\pi$ except that the ax-edge x–x' and the cut-edge x'–x'' are both removed, and that each ax-edge of endpoint x'' (corresponding to an ax-rule in τ) instead has endpoint x. See Figure 3 for an illustration of these two graphs.

We obtain a path p' between u and v in $\mathcal{G}_\pi$ by replacing in p any ax-edge y–x that does not belong to $\mathcal{G}_\pi$ by the edges y–x''–x'–x. Observe that p' is an alternating path since p is.

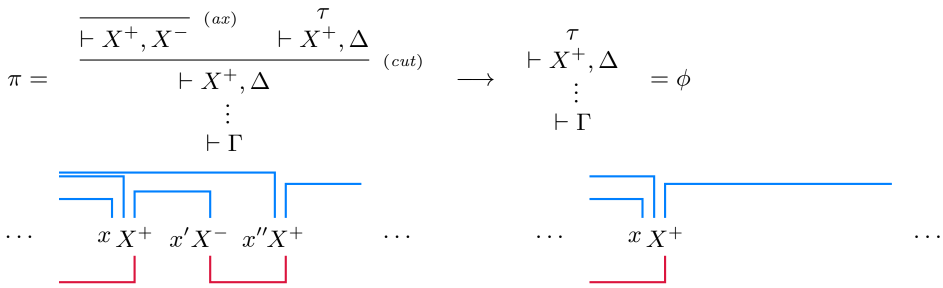

Figure 3 Illustration of the *ax* key case in the proof of Lemma 9

$?c - !$ **key case.** We have

$$\pi = \cfrac{\cfrac{\overset{\tau}{\vdash ?A^\perp, ?A^\perp, \Delta}}{\vdash ?A^\perp, \Delta}\,(?c) \qquad \cfrac{\overset{\mu}{\vdash A, ?\Sigma}}{\vdash !A, ?\Sigma}\,(!)}{\cfrac{\vdash \Delta, ?\Sigma}{\cfrac{\vdots}{\vdash \Gamma}}}\,(cut) \quad \longrightarrow \quad \cfrac{\cfrac{\overset{\tau}{\vdash ?A^\perp, ?A^\perp, \Delta} \qquad \cfrac{\overset{\mu}{\vdash A, ?\Sigma}}{\vdash !A, ?\Sigma}\,(!)}{\vdash ?A^\perp, \Delta, ?\Sigma}\,(cut) \qquad \cfrac{\overset{\mu}{\vdash A, ?\Sigma}}{\vdash !A, ?\Sigma}\,(!)}{\cfrac{\vdash \Delta, ?\Sigma, ?\Sigma}{\cfrac{\vdash \Delta, ?\Sigma}{\cfrac{\vdots}{\vdash \Gamma}}}\,(?c)}\,(cut) = \phi$$

with an alternating path p between u and v in the GOI projection graph $\mathcal{G}_\phi$. Let us call S the set of vertices in $\mathcal{G}_\pi$ corresponding to the atoms occurring in the *cut*-formula $!A$ and $S^\perp$ the set of vertices in $\mathcal{G}_\pi$ corresponding to the atoms occurring in the *cut*-formula $?A^\perp$, such that there is a *cut*-edge between $x \in S$ and $x^\perp \in S^\perp$.

We observe that the vertices of $\mathcal{G}_\phi$ are those of $\mathcal{G}_\pi$ where S is replaced by two of its copies, that we call $S_1 = \{x_1 \mid x \in S\}$ and $S_2 = \{x_2 \mid x \in S\}$, and similarly for $S^\perp$ with $S_1^\perp = \{x_1^\perp \mid x^\perp \in S\}$ and $S_2^\perp = \{x_2^\perp \mid x^\perp \in S\}$. Furthermore, the edges of $\mathcal{G}_\phi$ are those of $\mathcal{G}_\pi$ except that:

- each *ax*-edge y–$x^\perp$ coming from the sub-derivation τ with an endpoint $x^\perp \in S^\perp$ is replaced by an *ax*-edge y–$x_1^\perp$ or by an *ax*-edge y–$x_2^\perp$ (or possibly by both);
- each *ax*-edge y–x coming from the sub-derivation μ with an endpoint $x \in S$ is replaced by two *ax*-edges y–x_1 and y–x_2;
- instead of each *cut*-edge $x^\perp$–x, we have two *cut*-edges $x_1^\perp$–x_1 and $x_2^\perp$–x_2.

See Figure 4 for an illustration of these two graphs.

We obtain a path p' between u and v in $\mathcal{G}_\pi$ by replacing in p each occurrence of a $x_1 \in S_1$ and of a $x_2 \in S_2$ by $x \in S$, and similarly of a $x_1^\perp \in S_1^\perp$ and of a $x_2^\perp \in S_2^\perp$ by $x^\perp \in S^\perp$. That p' is an alternating path follows from the fact p is such a path. ◀

D Proof of Proposition 23 with Intermediate Results

We present here a proof of Proposition 23. It uses two intermediate results, the second one being the main novelty with respect to the proof of Proposition 14 in Section 4.

▶ **Lemma 28.** *Let F and C be formulas, and X and Y be atoms such that $X \notin \mathcal{V}(F)$ and $X \notin \mathcal{V}(C)$. Set:*

$$A \stackrel{def}{=} X^+ \otimes !\big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

If $\vdash A^\perp, C$ is provable, then there exists a cut-free derivation of $\vdash A^\perp, C$ in which no !-rule has an atom X in its conclusion sequent.

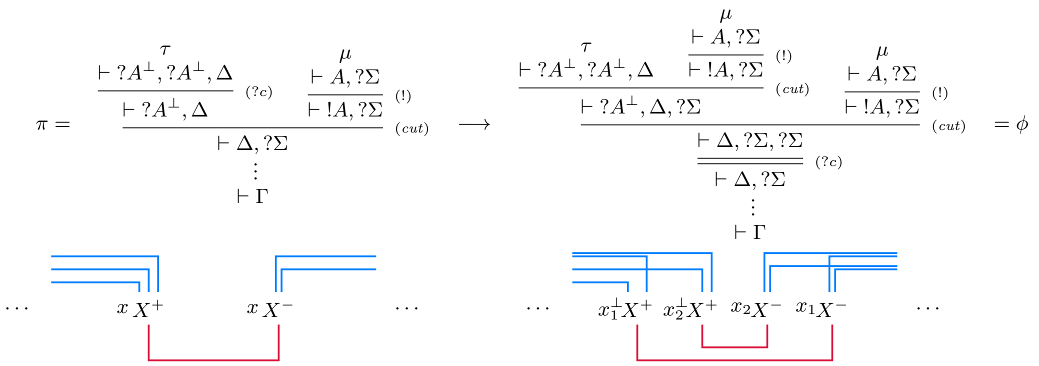

Figure 4 Illustration of the $?c - !$ key case in the proof of Lemma 9

Proof. Proceed exactly as in the proof of Lemma 12. ◀

▶ **Lemma 29.** *Let F and C be formulas, and X and Y be atoms such that $X, Y \notin \mathcal{V}(F)$ and $X \notin \mathcal{V}(C)$. Assume that F is not provable, and set:*

$$A \stackrel{def}{=} X^+ \otimes ! \left(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\right) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

$$\begin{cases} B_0 & \stackrel{def}{=} (Y^+ \multimap (Y^+ \otimes !F)) \otimes (Y^+ \multimap (Y^+ \otimes !F)) \\ B_{n+1} & \stackrel{def}{=} B_n \otimes (Y^+ \multimap (Y^+ \otimes !F)) \end{cases}$$

Let π be a cut-free derivation of $\vdash A^\perp, B_n$ for some $n \in \mathbb{N}$. Then in π there is no ax-rule on $\vdash Y^+, Y^-$ where both of these occurrences belong to B_n.

Proof. We proceed by contradiction. Such an *ax*-rule can only be between the occurrences Y^+ and Y^- of a sub-formula $Y^+ \multimap (Y^+ \otimes !F) = Y^- \parr (Y^+ \otimes !F)$, since other occurrences of Y in B_n are separated by a $\otimes$.

By the sub-formula property and simple (non-)provability results, this implies the following is a sub-tree of π:

$$\dfrac{\dfrac{\dfrac{}{\vdash Y^-, Y^+}\,(ax)}{\vdash Y^-, Y^+, ?\Delta}\,(?w) \qquad \begin{matrix}\phi \\ \vdash !F, \Gamma\end{matrix}}{\vdash Y^-, Y^+ \otimes !F, \Gamma, ?\Delta}\,(\otimes)$$

where the doubled inference rule means we apply the rule a certain number of times, and where the formulas of Γ are included in
$\{X^-, ?\left(X^+ \otimes (X^- \parr (Y^+ \otimes (Y^- \parr ?F^\perp)))\right), X^+ \otimes (X^- \parr (Y^+ \otimes (Y^- \parr ?F^\perp)), Y^+ \otimes (Y^- \parr ?F))\}$
with possibly some of these formulas linked by $\parr$ and formulas appearing possibly several times. We substitute Y^+ by 0 in ϕ, obtaining a derivation τ of a sequent (isomorphic to) $\vdash !F, \Gamma'$ where the formulas of Γ' are included in $\{X^-, ?\left(X^+ \otimes (X^- \parr 0)\right), X^+ \otimes (X^- \parr 0), 0\}$ with possibly some of these formulas linked by $\parr$ and formulas appearing possibly several times.

Such a derivation τ must contain a $!$-rule corresponding to the root connective of $!F$. By the sub-formula property, this means $\vdash !F, \Gamma''$ is provable with Γ'' composed of the formula $?\left(X^+ \otimes (X^- \parr 0)\right)$ a certain number of times. Now, substituting X^+ by 0, we obtain a derivation of conclusion $\vdash !F, \Gamma'''$ with Γ'' composed of a formula isomorphic to $?0$ a certain number of times. From this last derivation, we deduce that $\vdash F$ is provable, a contradiction. Indeed, if the formula $?0$ appears at least once we have the derivation

$$\dfrac{\dfrac{\vdash !F, ?0, \cdots, ?0}{\vdash !F, ?0}\ (?c) \qquad \dfrac{\dfrac{}{\vdash \top}\ (\top)}{\vdash !\top}\ (!)}{\vdash !F}\ (cut)$$

and if it does not appear we directly have a derivation of $!F$. But the last rule of a *cut*-free derivation of $\vdash !F$ can only be $\dfrac{\vdash F}{\vdash !F}\ (!)$, so F is provable and we are done. ◀

▶ **Proposition 23.** *Let F and C be formulas, and X and Y be atoms such that $X, Y \notin \mathcal{V}(F)$ and $X \notin \mathcal{V}(C)$. Assume that F is not provable, and set:*

$$A \stackrel{def}{=} X^+ \otimes !\big(X^+ \multimap (X^+ \otimes (Y^+ \multimap (Y^+ \otimes !F)))\big) \otimes (X^+ \multimap (Y^+ \multimap (Y^+ \otimes !F)))$$

$$\begin{cases} B_0 & \stackrel{def}{=} (Y^+ \multimap (Y^+ \otimes !F)) \otimes (Y^+ \multimap (Y^+ \otimes !F)) \\ B_{n+1} & \stackrel{def}{=} B_n \otimes (Y^+ \multimap (Y^+ \otimes !F)) \end{cases}$$

If $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$ are both provable for some number $n \in \mathbb{N}$, then C contains at least $n+2$ occurrences of Y^+.

Proof. We proceed as in the proof of Proposition 14. The differences with that proof are **in bold and colored in green**.

Let us call π and ϕ derivations of respectively $\vdash A^\perp, C$ and $\vdash C^\perp, B_n$. We assume them *cut*-free without any loss of generality. We will use the GOI projection graph of (some 0-slice of) their composition by *cut* to identify Y^-'s in $C^\perp$ linked to the Y^+'s of B_n. Thanks to alternating paths, we will prove that these occurrences of Y^- are pairwise distinct. This yields one Y^- in $C^\perp$ for each Y^+ in B_n, concluding the proof.

Finding an alternating path in some 0-slice. Observe that $\dfrac{\overset{\pi}{\vdash A^\perp, C} \qquad \overset{\phi}{\vdash C^\perp, B_n}}{\vdash A^\perp, B_n}\ (cut)$ reduces to a *cut*-free derivation of $\vdash A^\perp, B_n$, that we call τ. **Consider any 0-slice τ' of τ.** By repeated applications of Lemma 4, there is a 0-slice μ of $\dfrac{\overset{\pi}{\vdash A^\perp, C} \qquad \overset{\phi}{\vdash C^\perp, B_n}}{\vdash A^\perp, B_n}\ (cut)$ such that $\mu \longrightarrow^* \tau'$.

By Lemma 29, there is no ax-rule on $\vdash Y^+, Y^-$ where both of these occurrences belong to B_n. Consider some occurrence Y^+_α in B_n, with the index α used to distinguish it from other occurrences. Since τ' is a *cut*-free **0-slice** of $\vdash A^\perp, B_n$, there must be an ax-rule between this occurrence Y^+_α and one of the two occurrences of Y^- in $A^\perp$: call it Y^-_1. Hence, there is an alternating path (made of one ax-edge) between these two occurrences in the GOI projection graph $\mathcal{G}_\tau$ of τ. By repeated applications of Lemma 9, there also is an alternating path p between Y^-_1 and Y^+_α in $\mathcal{G}_\mu$. Thus, there is an occurrence Y^-_β and an ax-rule $\dfrac{}{\vdash Y^-_\beta, Y^+_\alpha}\ (ax)$ in μ and Y^-_β–Y^+_α is the last edge of p. This occurrence Y^-_β can only be in $C^\perp$.

We use the alternating path p to show there is no sub-formula $?D$ of $C^\perp$ containing Y^-_β – say differently, there is no $?$ between the root of the formula $C^\perp$ and Y^-_β. Once this claim is proved, we are done: each occurrence of Y^+ in B_n is linked by an ax-edge to an occurrence of Y^- in $C^\perp$. All these occurrences of Y^- are pairwise distinct (recall Remark 8): there is no sub-formula $?D$ of $C^\perp$ containing such an occurrence, so they cannot be contracted; and there is no &-rule that can share the identified ax-rules since we are in a 0-slice, and these ax-rules cannot be above a !-rule for there is no ?-connective nor !-connective below the Y^+ of B_n. Therefore, we have found $n+2$ occurrences of Y^- in $C^\perp$, as wished.

Exploiting the alternating path. We prove our claim by showing a stronger result:

- for all occurrences of Y^- in C (respectively in $C^\perp$) belonging to p, there is no ? between the root of C (respectively of $C^\perp$) and these occurrences;
- for all occurrences of Y^+ in C (respectively in $C^\perp$) belonging to p, there is no ! between the root of C (respectively of $C^\perp$) and these occurrences.

We proceed by following the edges of p starting from its endpoint Y_1^- in $A^\perp$.

Exploiting the alternating path: base case. Consider the first edge $Y_1^-–Y_2^+$ of p, with Y_2^+ necessarily in C. Using **Lemma 28**, without any loss of generality, there is no !-connective between the root of C and Y_2^+. Otherwise, since there is an ax-rule on $\vdash Y_1^-, Y_2^+$, there would be a !-rule (associated to this !-connective) with in its conclusion sequent a ?-sub-formula of $A^\perp$ containing Y_1^-, meaning with $?(X^+ \otimes (X^- \parr Y^-))$: contradiction.

Exploiting the alternating path: inductive case. See Figure 2 for an illustration of our present reasoning. Assume the results holds for a prefix of p, and let us consider its last edge.

1. Suppose this prefix ends with an ax-edge $Y_k^-–Y_{k+1}^+$ with Y_{k+1}^+ in C. We know there is no ! between the root of C and Y_{k+1}^+. As p is alternating, the subsequent edge is a cut-edge $Y_{k+1}^+–Y_{k+2}^-$ with Y_{k+2}^- in $C^\perp$. Since Y_{k+1}^+ and Y_{k+2}^- are dual occurrences by definition of a cut-edge, there is no ? between the root of $C^\perp$ and Y_{k+2}^-.
2. Suppose this prefix ends with a cut-edge $Y_k^+–Y_{k+1}^-$ with Y_{k+1}^- in $C^\perp$. We know there is no ? between the root of $C^\perp$ and Y_{k+1}^-. As p is alternating, the subsequent edge is an ax-edge $Y_{k+1}^-–Y_{k+2}^+$ with Y_{k+2}^+ either in B_n or in $C^\perp$. In the first case, we are done: we have $Y_{k+2}^+ = Y_\alpha^+$ since there is no cut-edge with endpoint Y_{k+2}^+ to continue our path, and p is supposed to end in Y_α^+. In the second case, since there is an ax-rule $\dfrac{}{\vdash Y_{k+1}^-, Y_{k+2}^+}\,(ax)$ by definition of an ax-edge, it follows there is no ! between the root of $C^\perp$ and Y_{k+2}^+ – because the context condition of a corresponding !-rule would prevent the presence of Y_{k+1}^- in all sequents above it.
3. Suppose this prefix ends with an ax-edge $Y_k^-–Y_{k+1}^+$ with Y_{k+1}^+ in $C^\perp$. This case is identical to case 1 up to switching C and $C^\perp$.
4. Suppose this prefix ends with a cut-edge $Y_k^+–Y_{k+1}^-$ with Y_{k+1}^- in C. This case is identical to case 2, with no need to consider Y_{k+2}^+ in $A^\perp$ since $A^\perp$ has no Y^+.

This proves our claim, and concludes the proof. ◀